

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.7_1)

Metric

$$ds^2 = (-x^2 B(u) - y^2 B(u)) du \otimes du - du \otimes dv - dv \otimes du + dx \otimes dx + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{1}{2} (x^2 + y^2) B(u) \right) du + (1) dv$$

$$m = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(\frac{1}{2} i \sqrt{2} \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} \right) dx + \left(-\frac{1}{2} i \sqrt{2} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(\frac{1}{2} (x^2 + y^2) B(u) \right) dv$$

$$m = \left(\frac{1}{2} \sqrt{2}\right) dx + \left(\frac{1}{2} i \sqrt{2}\right) dy$$

$$\overline{m} = \left(\frac{1}{2} \sqrt{2}\right) dx + \left(-\frac{1}{2} i \sqrt{2}\right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + \frac{1}{2} (x^2 + y^2) B(u) \partial_v$$

$$\delta = \frac{1}{2} \sqrt{2} \partial_x + \frac{1}{2} i \sqrt{2} \partial_y$$

$$\bar{\delta} = \frac{1}{2} \sqrt{2} \partial_x - \frac{1}{2} i \sqrt{2} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = \frac{1}{2} \sqrt{2} (x - i y) B(u)$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = 0$$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$\Phi_{22} = B(u)$

$\Lambda = 0$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$\Psi_2 = 0$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"